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NANYANG TECHNOLOGICAL UNIVERSITY

SEMESTER 1 EXAMINATION 2025-2026

IE2104 – DIGITAL ELECTRONICS

November / December 2025 Time Allowed: 2 hours

INSTRUCTIONS

1. This paper contains 4 questions and comprises 5 pages.
2. Answer all questions.
3. All questions carry equal marks.
4. This is a closed book examination.
5. Unless specifically stated, all symbols have their usual meanings.

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1. (a) A 2's complement number expressed in the octal representation is converted to decimal value of 423 when it is mistaken as an unsigned number. What is the correct decimal value of the 2's complement number?

(8 Marks)

(b) Given the logic function $y(a,b,c,d) = a \cdot (d \cdot \bar{b} + c)$, draw a CMOS circuit with minimum number of transistors for the function.

(7 Marks)

(c) Given the logic function $f(a,b,c,d) = \Pi M(2,5,7,10,11) + x(3,4,9,13,14)$, where $x(3,4,9,13,14)$ are don't care terms, find:

(i) the minimized sum of products of $f(a,b,c,d)$;

(ii) the minimized product of sums of $f(a,b,c,d)$.

(10 Marks)

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2. (a) Implement the logic function $f(a,b,c) = \Pi M(1,3,4,5,7)$ with the decoder given in Table 1 and a logic gate.

(6 Marks)

Table 1

Input			Output							
I_2	I_1	I_0	Y_7	Y_6	Y_5	Y_4	Y_3	Y_2	Y_1	Y_0
0	0	0	1	1	1	1	1	1	1	0
0	0	1	1	1	1	1	1	1	0	1
0	1	0	1	1	1	1	1	0	1	1
0	1	1	1	1	1	1	0	1	1	1
1	0	0	1	1	1	0	1	1	1	1
1	0	1	1	1	0	1	1	1	1	1
1	1	0	1	0	1	1	1	1	1	1
1	1	1	0	1	1	1	1	1	1	1

(b) Implement the logic function $g(a,b,c) = (\bar{b}+c)\cdot(\overline{b \oplus \bar{a}})$ with the multiplexer given in Table 2.

(7 Marks)

Table 2

Select		Output
S_1	S_0	Y
0	0	D_0
0	1	D_1
1	0	D_2
1	1	D_3

(c) Derive the maxterm expression in the form of $\Pi M(...)$ of the logic function *h(x, y, z)* for the logic circuit in Figure 1 on page 3, where the function table of the MUX is given in Table 2 and the labels of the full adder have their usual meanings.

(7 Marks)

Note: Question No. 2 continues on page 3.

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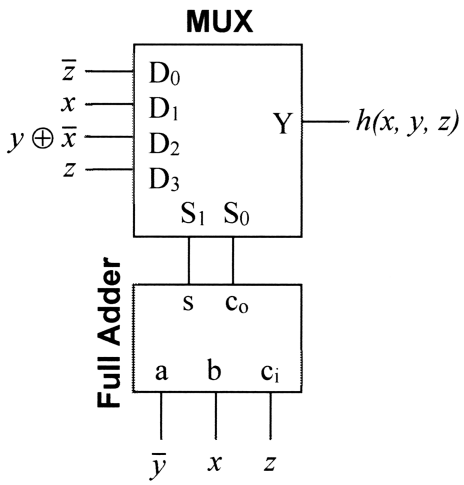


Figure 1

(d) Given the logic circuit in Figure 2, where $X = x_{n-1}x_{n-2}...x_0$ and $Y = y_{n-1}y_{n-2}...y_0$ are 2 n-bit binary numbers, and the logic function of the blocks is $k(a, b, c) = (a + b + c) \cdot (\overline{a} + \overline{b} + c)$. In one short sentence, state the function of the output z.

(5 Marks)

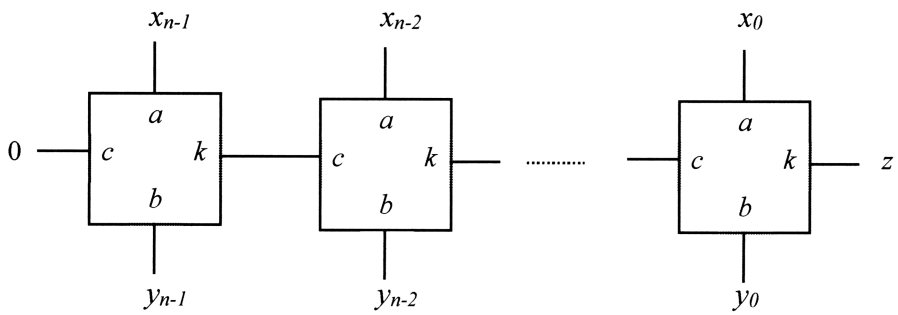


Figure 2

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3. (a) Figure 3 shows a sequential circuit, where *X* and *Z* are the sequence input and output, respectively, *Clk* is the clock signal, and *Rst* is an active low reset signal to initialize the sequential circuit.

(i) Is this circuit a Moore or a Mealy state machine? State your reason.

(ii) Write the input functions of the T and D flip-flops.

(iii) Construct the state transition table.

(iv) Draw the state diagram.

(v) What bit patterns can this circuit detect? Are the detected patterns overlapping or non-overlapping?

(18 Marks)

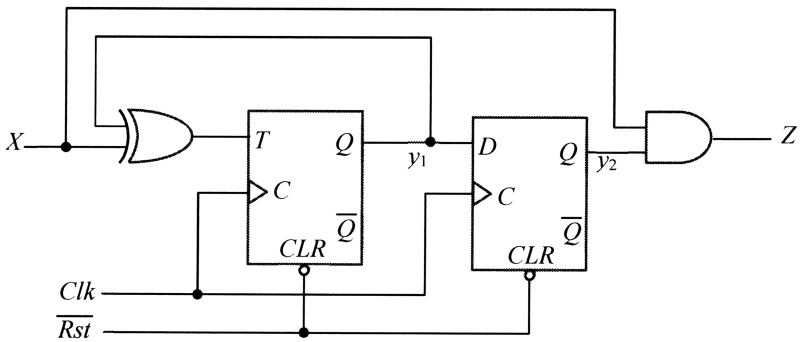


Figure 3

(b) The device timing parameters (in ns) for Figure 3 are given in Table 3.

(i) Determine the setup and hold time margins if the frequency of *Clk* is 25 MHz.

(ii) What is the maximum clock frequency for the circuit to operate without setup and hold time violations?

(7 Marks)

Table 3

Device	t_d (min)	t_d (max)	t_{su} (setup)	t_h (hold)
XOR	6	12	-	-
AND	4	6	-	-
T flip-flop	4	10	7	5
D flip-flop	3	6	5	4

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4. (a) A standard Gray code sequence for the integers 0 to 7 is 000, 001, 011, 010, 110, 111, 101, 100. Design a synchronous 3-bit bidirectional Gray code sequence counter using T flip-flops and logic gates. The counter has a control input M. The output $Q_2Q_1Q_0$ represents the current count value, with Q_2 being the most significant bit and Q_0 the least significant bit. The counter should count up when M is 0 and count down when M is 1.

(i) Draw the state diagram of this counter.
(ii) Construct the state transition table.
(iii) Draw K-maps to determine the flip-flop input equations. Simplify the next state decoder logic with multi-input XOR/XNOR and AND gates if their use can reduce the gate count compared to sum-of-products or product-of-sums forms.
(iv) Draw the final circuit.

(20 Marks)

(b) State the pros and cons of a synchronous counter compared to an asynchronous counter.

(5 Marks)

END OF PAPER

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IE2104 DIGITAL ELECTRONICS

Please read the following instructions carefully:

1. **Please do not turn over the question paper until you are told to do so. Disciplinary action may be taken against you if you do so.**

2. You are not allowed to leave the examination hall unless accompanied by an invigilator. You may raise your hand if you need to communicate with the invigilator.

3. Please write your Matriculation Number on the front of the answer book.

4. Please indicate clearly in the answer book (at the appropriate place) if you are continuing the answer to a question elsewhere in the book.